

University Interscholastic League
2023 – 2024 Junior High Number Sense Test B

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.
Stop – Wait for Signal!

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| <p>(1) $23 + 102 =$ _____</p> <p>(2) $14 \times 4 =$ _____</p> <p>(3) $74 - 31 =$ _____</p> <p>(4) $248 \div 4 =$ _____</p> <p>(5) $\frac{7}{9} - \frac{2}{3} =$ _____</p> <p>(6) $10 \div 0.5 =$ _____</p> <p>(7) $11 \times 33 =$ _____</p> <p>(8) $11^2 =$ _____</p> <p>(9) $12 + 12 \div 6 =$ _____</p> <p>*(10) $223 + 2023 + 2024 =$ _____</p> <p>(11) $31 \times 29 =$ _____</p> <p>(12) If the arithmetic mean of x, 12 and 20 is 11, $x =$ _____</p> <p>(13) Which is smaller: $\frac{5}{12}$ or $\frac{3}{7}$? _____</p> <p>(14) $9 \times 15 - 5 \times 15 =$ _____</p> <p>(15) $24 \times 84 =$ _____</p> <p>(16) $59 - 16 - 14 =$ _____</p> <p>(17) $44^2 =$ _____</p> <p>(18) $11 + 15 + 19 + 23 =$ _____</p> <p>(19) MMXXIII = _____ (Arabic Numeral)</p> | <p>*(20) $187.5 \times 160 =$ _____</p> <p>(21) $39 \times 41 + 1 =$ _____</p> <p>(22) $234 \times 101 =$ _____</p> <p>(23) If $f(x) = -2x^2 + 10$ then $f(-1) =$ _____</p> <p>(24) $8\frac{1}{10} \times 2\frac{1}{10} =$ _____ (mixed number)</p> <p>(25) $2\frac{5}{6} + 3\frac{2}{3} =$ _____</p> <p>(26) $15 - 15^2 =$ _____</p> <p>(27) The negative square root of 361 is _____</p> <p>(28) If n is to 12 as 5 is to 6, then $n =$ _____</p> <p>(29) If $11 + 5x$ is 36, then $x =$ _____</p> <p>*(30) $310^2 =$ _____</p> <p>(31) $52 \times 58 =$ _____</p> <p>(32) The sum of the prime numbers greater than 0 but less than 11 is _____</p> <p>(33) If $2.25 - 0.5 = n$, the $n^{-1} =$ _____</p> <p>(34) 18 is divisible by how many whole numbers? _____</p> <p>(35) 2 square feet = _____ square inches</p> <p>(36) 8 percent = _____ (common fraction)</p> <p>(37) If the sales tax is $8\frac{1}{2}\%$, what is the amount of sales tax on an item that costs \$20? \$ _____</p> |
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- (38) The LCM (12, 16) $\times$ GCF (12, 16) = _____
- (39) The area of an equilateral triangle with a side of 2-cm is $a\sqrt{3}$ and a = _____ cm²
- *(40) $111\sqrt{19600}$ = _____
- (41) 34 (base 5) = _____ (base 10)
- (42) $4^3 - 3^2$ = _____
- (43) 58×111 = _____
- (44) What is the area of a rhombus with diagonals 64 and 60? _____
- (45) 42 (base 6) $- 5$ (base 6) = _____ (base 6)
- (46) If $24 \div 6 = n^2 - 12$, and $n > 0$, then n = _____
- (47) If $A = \{1, 2, 3, 4\}$, $B = \{0, 1, 2, 3\}$ and $C = \{4, 2, 0\}$, how many elements are in $A \cap C \cap B$? _____
- (48) 24 pints = _____ quarts
- (49) The surface area of a rectangular solid that measures 6 by 1 by 4 is _____
- *(50) 240×624 = _____
- (51) In the sequence: 2, 4, a , 8, 10, b , 14, . . .
 $(b - a)^2$ = _____
- (52) What is the 6th term in the sequence: 1, 3, 5, . . .? _____
- (53) Each exterior angle for a square is _____ $^\circ$
- (54) What is the area of a parallelogram with opposite sides each 10 and altitude between them 25? _____
- (55) What is the area of a rectangle with a perimeter of 30 and length 9? _____
- (56) What is the radius of a circle with a circumference of 64π ? _____
- (57) The hypotenuse for a triangle with sides 10, 24 is _____
- (58) $(23^2 + 15 \times 4) \div 5$ has a remainder of _____
- (59) If the surface area for a cube is 216, its edge is _____
- *(60) $24^4 \times 101 \div 4^3$ = _____
- (61) $(5^{-1} \times 5^2)^{-1}$ = _____
- (62) 366 feet = _____ yards
- (63) 18 km/hr = _____ meters/sec
- (64) $16\frac{2}{3}\% + 44\frac{4}{9}\%$ = _____ (common fraction)
- (65) $15^3 \div 6$ has a remainder of _____
- (66) $\frac{1+2+3+\dots+6}{2+4+6+\dots+10}$ = _____
- (67) The sum of the positive divisors of 12 is _____
- (68) $5!$ = _____
- (69) $4\frac{1}{2} \div \frac{3}{8}$ = _____
- *(70) 11 miles = _____ yards
- (71) $12^2 + 24^2$ = _____
- (72) $45\frac{5}{11}\%$ = _____ (common fraction)
- (73) $0.0333\dots$ = _____ (common fraction)
- (74) The volume of a sphere with a diameter of 6 is $k\pi$ and k = _____
- (75) What are the odds of randomly picking a blue ball from a bag of 20 red and 18 blue balls? _____
- (76) $12^2 \times 25$ = _____
- (77) If the shortest distance between the points (15, 0) and (0, x) is 17, what is x ? _____
- (78) $11 \times \frac{6!}{4!}$ = _____
- (79) $(3.5)^2 - (7.5)^2$ = _____
- *(80) 750 hours = _____ minutes

2023 – 2024 University Interscholastic League Junior High Number Sense Test B – Key

(1) 125	*(20) 28500 – 31500	(38) 192	(59) 6
(2) 56	(21) 1600	(39) 1	*(60) 497405 – 549763
(3) 43	(22) 23634	*(40) 14763 – 16317	(61) $\frac{1}{5}$; .2
(4) 62	(23) 8	(41) 19	(62) 122
(5) $\frac{1}{9}$	(24) $17\frac{1}{100}$	(42) 55	(63) 5
(6) 20	(25) $6\frac{1}{2}$; 6.5; $\frac{13}{2}$	(44) 1920	(64) $\frac{11}{18}$
(7) 363	(26) -210	(45) 33	(65) 3
(8) 121	(27) -19	(46) 4	(66) $\frac{7}{10}$; .7
(9) 14	(28) 10	(47) 1	(67) 28
*(10) 4057 – 4483	(29) 5	(48) 12	(68) 120
(11) 899	*(30) 91295 – 100905	*(49) 68	(69) 12
(12) 1	(31) 3016	*(50) 142272 – 157248	*(70) 18392 – 20328
(13) $\frac{5}{12}$	(32) 17	(51) 36	(71) 720
(14) 60	(33) $\frac{4}{7}$	(52) 11	(72) $\frac{5}{11}$
(15) 2016	(34) 6	(53) 270	(73) $\frac{1}{30}$
(16) 29	(35) 288	(54) 250	(74) 36
(17) 1936	(36) $\frac{2}{25}$	(55) 54	(75) $\frac{9}{10}$; .9
(18) 68	(37) 1.70	(56) 32	(76) 3600
(19) 2023		(57) 26	(77) 8
		(58) 4	(78) 330
			(79) -44
			*(80) 42750 – 47250

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.